

Pivot Slide Model – Summary

Model Idea:

The stone pivots around a pebble in the ice, the pivot point is the point on the running band that is touching the pebble.

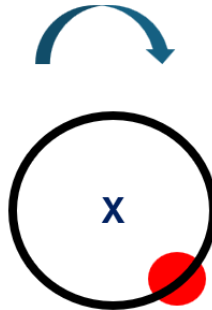


Figure 1: The running band of the curling stone, the X represents the stone centre and the red circle is the pebble the running band pivots around.

It only pivots for a short time before continuing to slide, this time is called the pivot time and can be calculated using the equation $t = \frac{t_{max}}{v}$ where t is the pivot time, t_{max} is the maximum pivot time with a value of 45ns and v is magnitude of the velocity.

When pivoting, to find the angular velocity, the equation $\omega = v/r_{pivot}$ is used, where ω is the angular velocity of the pivot, v is the velocity of the stone before the pivot and r is the radius of the pivot.

The stone is either pivoting or sliding at any point in time.

Lots of small pivots add up over time to make the curling stone curl.

I am now assuming that the friction coefficient is not a constant and can be determined using the following equation: $\mu = 0.025 \times v^{-0.5}$ where μ is the coefficient of friction and v is the translational velocity of the curling stone.

The equation for the coefficient of friction was adapted from (Kietzig et al., 2010).

The equations used for pivot time, pivot angular velocity were taken from (Shegelski et al., 2018).

Optimisation:

The way the code was written had to be adapted so that it was more efficient and could be completed within a practical time.

The method check grid cell was created to check the location of the stone and determine whether it has moved from one grid cell to another. If it has moved grid cells, a list of pebbles, that stores pebbles in the same grid cell as the stone and the neighbouring cells, is updated and these are called the relevant pebbles. This means that instead of checking thousands of pebbles every timestep, only pebbles that are close to the stone are checked.

The other thing improved was two different timesteps depending on whether the stone is pivoting or sliding. A smaller timestep is required for pivoting motion however a larger timestep can be used while sliding.

Assumptions:

Ignoring the height of the pebbles.

Constant coefficient of friction for the rotation of the stone.

Predictions:

Predicts that the stone will curl in the same direction as the stone's rotation.

Curling stone should pivot around at least 500 pebbles over 25 metres.

Each run should produce different y displacements due to the pebbles being randomly distributed on the ice.

The curl should be in the same direction as the rotation of the curling stone.

Results:

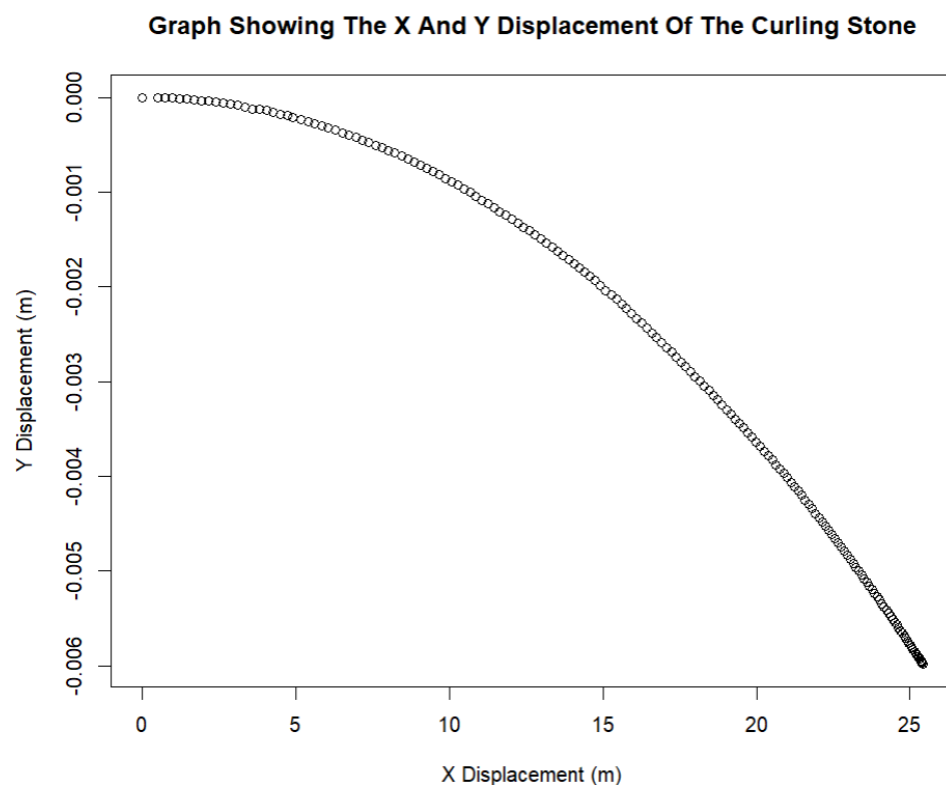


Figure 2: The motion of the curling stone with a clockwise rotation during one specific run, shows a curl of about 6mm

Over multiple runs, the average y displacement for this model is 6.564mm with a standard deviation of 0.006564 and the average number of pivots per run is 782 with a standard deviation of 42.5245.

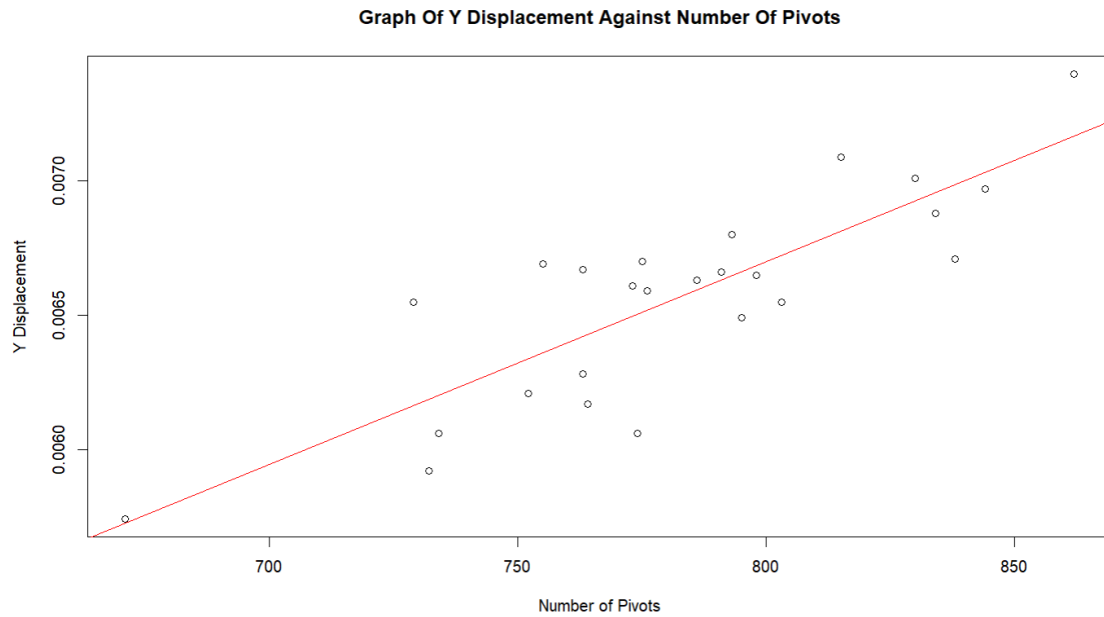


Figure 3: The relationship between the Y displacement and the number of times the stone pivots in one run, this graph was made after 25 runs of the curling stone, and this data was also used to create the averages stated earlier

The above shows a weak positive correlation between the number of times the stone pivots and the magnitude of the curl it reaches. The reason that this is not a strong positive correlation is that a pivot nearer the end of the stone's path produces a larger y displacement than a pivot at the start. It is a positive correlation as each pivot increases the y displacement.

Conclusion:

Overall, this model produces an average y displacement of 6.564mm and an average number of pivots of 782 pivots. The number of pivots is in the same order of magnitude as expected while the y displacement is 2 orders of magnitude lower than expected. Therefore, this model is not producing the magnitude of curl that we see in real life.

Future Work:

- Accounting for the pebble height
- Variable friction on the angular velocity
- Combine with other models to achieve a larger y displacement